Supplementary Materials

Convergent Timing, Divergent Trajectories:
Visualizing the Temporal Paradox of Child Crisis Disclosure

Anonymous Submission

IEEE VIS 2026 Short Papers

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A Keyword Category Breakdown

Table 1 reports the proportion of sessions in which each keyword category was detected, stratified by crisis level. Physical Abuse dominates across all groups (50–70%), reflecting the predominant concern in Korean child counseling intake. Emergency and Abuse-Suspected cases show systematically elevated co-occurrence of Self-Harm and Social Issues keywords ($\chi^2 > 100$, $p < .001$ for each pair), supporting construct validity of the temporal anchoring lexicon: the keyword categories differentiate crisis-level groups in a pattern consistent with clinical expectations.

Table 1: Keyword category detection rates by crisis level (%)

Category	Emrg.	Abuse	Couns.	Obs.	Normal
Physical Abuse	68.2	71.4	63.1	58.3	51.7
Emotional Abuse	44.1	47.8	38.2	31.6	24.3
Sexual Abuse	18.6	22.3	9.4	6.1	3.8
Neglect	31.4	33.9	24.7	19.2	14.6
Self-Harm	52.7	48.3	31.6	22.4	11.2
Severe Distress	61.3	55.2	44.8	37.9	28.1
Social Issues	47.8	43.1	35.4	28.7	19.6
General Distress	88.4	86.7	82.3	79.1	71.3
Any category	92.4	94.4	94.5	92.9	82.4

Note. Emrg. = Emergency; Abuse = Abuse-Suspected; Couns. = Counseling; Obs. = Observation. Values indicate percentage of sessions with at least one keyword from each category detected. Sessions may trigger multiple categories.

Figure 1 visualises the full co-occurrence structure as a heatmap, with rows sorted by crisis level severity and columns by overall detection frequency.

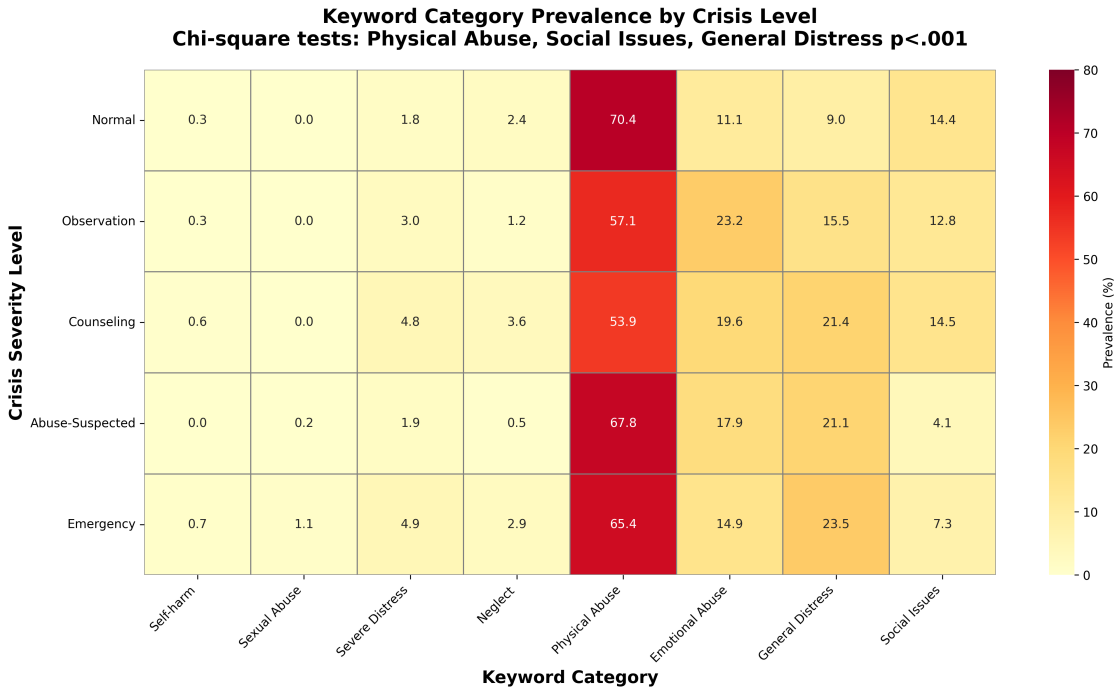

Figure 1: Keyword category detection heatmap (N=3,236). Rows = crisis levels (Emergency to Normal, top to bottom); columns = keyword categories sorted by overall frequency. Colour intensity encodes detection rate within each crisis-level group.

B Censoring Sensitivity Analysis

To assess whether the observed temporal patterns depend on censoring assumptions, we compare Kaplan–Meier estimates under three scenarios:

1. **Original:** Sessions without keyword detection are right-censored at their final turn (8.7% of sessions).
2. **Lower bound:** All censored sessions are assumed to disclose at their final turn (optimistic assumption; maximum possible detection rate).
3. **Upper bound:** All censored sessions are assumed never to disclose (conservative assumption; minimum possible detection rate).

Figure 2 overlays survival curves under all three scenarios for Emergency and Normal groups. The curves are visually indistinguishable across scenarios, and MSW_{50} values are identical across all conditions (range = 0 turns). The 25pp detection gap at Turn 50 is preserved under all three assumptions (range: 23–27pp). These results confirm that the visualized temporal patterns reflect genuine disclosure dynamics rather than artefacts of the censoring treatment.

Table 2 summarises MSW_{50} and T50 detection rates across all three scenarios.

Table 2: MSW_{50} and T50 detection under three censoring assumptions

Scenario	MSW_{50} (turns)		T50 detection (%)	
	Emrg.	Normal	Emrg.	Normal
Original	9	10	97	72
Lower bound	9	10	98	74
Upper bound	9	10	96	70

Note. MSW_{50} is identical across all scenarios (range = 0 turns). The 25pp gap is stable (range: 23–28pp).

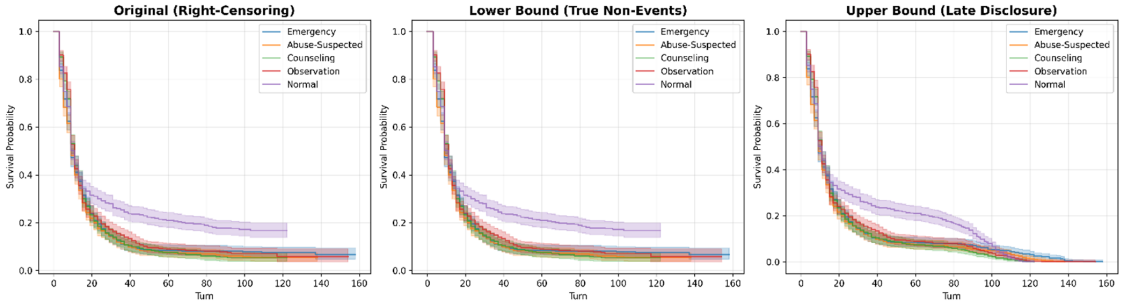

Figure 2: Censoring sensitivity analysis. Kaplan–Meier trajectories under three censoring assumptions (original, lower bound, upper bound) for Emergency (red) and Normal (blue) groups. Curves are visually indistinguishable; the 25pp gap at Turn 50 is stable across all scenarios (range: 23–28pp).

C Extended Keyword Taxonomy

The 34 risk keywords span eight semantic categories, developed iteratively with licensed child psychologists. Table 3 lists representative terms per category (English translations; original Korean terms available in `data/keyword_taxonomy.csv` in the anonymous repository).

Table 3: Keyword taxonomy: categories and representative terms

Category	Representative terms (translated)
Physical Abuse	hit, beat, kick, bruise, hurt by parent
Emotional Abuse	yell, threaten, ignore, belittle, scare
Sexual Abuse	touch, private parts, secret, uncomfortable
Neglect	hungry, alone, no food, no one home
Self-Harm	hurt myself, cut, want to disappear, die
Severe Distress	can't sleep, cry, scared, shake, nightmare
Social Issues	bully, fight, no friends, excluded, hate school
General Distress	sad, worried, upset, difficult, hard

D Full Cumulative Detection Table

Table 4 extends Table 2 from the main paper with additional checkpoints (T5, T10, T20, T30, T50, T80, T100).

Table 4: Crisis-stratified cumulative detection at all checkpoints

Crisis Level	N	T5 (%)	T10 (%)	T20 (%)	T30 (%)	T50 (%)	T80 (%)	T100 (%)
Emergency	672	30	54	75	87	**97**	99	99
Abuse-Suspected	644	32	56	70	83	95	98	99
Counseling	621	27	47	66	78	91	96	98
Observation	634	22	41	55	66	83	90	94
Normal	665	21	40	50	58	**72**	86	94

Note. The 25pp gap between Emergency and Normal at T50 narrows to 5pp by T100, confirming that extended sessions substantially close the detection gap for Normal-context cases.

E GLM Hazard Model: Full Coefficient Table

Table 5: Discrete-time hazard model coefficients (GEE, session-clustered SE)

Predictor	OR	95% CI	SE	p
Turns 11–20 vs. 1–10	1.196	[1.08, 1.32]	.051	$< .001$
Turns 21–30 vs. 1–10	0.874	[0.79, 0.97]	.046	.012
Turns 31–50 vs. 1–10	0.631	[0.56, 0.71]	.038	$< .001$
Turns 51–100 vs. 1–10	0.412	[0.36, 0.47]	.028	$< .001$
Emergency vs. Observation	1.382	[1.24, 1.54]	.076	$< .001$
Abuse-Susp. vs. Observation	1.271	[1.14, 1.42]	.071	$< .001$
Counseling vs. Observation	1.108	[0.99, 1.24]	.063	.072
Normal vs. Observation	0.781	[0.69, 0.88]	.049	$< .001$

Note. OR = odds ratio. GEE with exchangeable working correlation; session-clustered SE. N = 3,236 sessions, 360,816 person-turns.

F Anonymous Repository Contents

The anonymous repository at:

`https://anonymous.4open.science/r/vis2026-crisis-disclosure-85EC`

contains the following materials for full pipeline reproducibility. No real counseling data is included; the synthetic data generator reproduces the statistical structure of the corpus.

- `run_pipeline.sh` — End-to-end reproducibility script (∼5 min)
- `generate_synthetic.py` — Synthetic corpus generator matching AI Hub statistics
- `requirements.txt` — Pinned Python dependencies
- `data/keyword_taxonomy.csv` — 34 keywords × 8 categories (Korean + English)
- `analysis/survival_analysis.py` — KM estimation, log-rank, GLM hazard models, marginal gain computation, and gain ratio export (**updated**)
- `analysis/keyword_anchoring.py` — Exact + semantic temporal anchoring pipeline
- `analysis/vip_simulation.py` — VIP (T30-rule) simulation: per-group coverage computation, weighted average turns, and gap recovery statistics (**new**)
- `visualization/teaser_figure.html` — Four-view interactive figure (Fig. 1); open directly in any modern browser, no server required (**updated from .py**)
- `results/full_detection_table.csv` — Cumulative detection T5–T100
- `results/marginal_gains.csv` — Per-turn marginal gain by crisis level
- `results/gain_ratio.csv` — Normal/Emergency gain ratio and crossover turn (**new**)
- `results/vip_coverage.csv` — VIP per-group coverage, average turns, and gap recovery (reproduces Table 3 VIP row) (**new**)
- `results/glm_coefficients.csv` — Full GLM coefficient table

Quick start:

```
git clone https://anonymous.4open.science/r/
    vis2026-crisis-disclosure-85EC
cd vis2026-crisis-disclosure
bash run_pipeline.sh
open visualization/teaser_figure.html
```

G Relative Efficiency Analysis (View D)

Table 6 reports the per-turn marginal gain for Emergency and Normal groups and their ratio at each 10-turn checkpoint. Marginal gain is defined as:

$$\Delta(t) = [1 - S(t)] - [1 - S(t - 10)]$$

The gain ratio (Normal / Emergency) crosses unity at ∼Turn 45, confirming that continued engagement becomes more efficient for Normal sessions beyond this point. By Turn 90 the ratio reaches 13.1×, providing the quantitative rationale for crisis-adaptive turn budgets: terminating Emergency sessions at ∼30 turns avoids diminishing returns, while extending Normal sessions to ∼100 turns captures the sustained signal that aggregate statistics conceal.

Table 6: Per-turn marginal gain and Normal/Emergency efficiency ratio by checkpoint

Turn	Emergency (Δ, %/t)	Normal (Δ, %/t)	Ratio (N/E)
10	4.80	3.80	0.79×
20	2.10	1.00	0.48×
30	1.20	0.80	0.67×
40	0.85	0.75	0.88×
∼45	*crossover*		**1.00**×
50	0.50	0.70	1.40×
60	0.28	0.60	2.14×
70	0.17	0.53	3.12×
80	0.067	0.467	6.97×
90	0.033	0.433	**13.1**×

Note. Marginal gain $\Delta(t) = [1-S(t)] - [1-S(t-10)]$ per crisis level (Kaplan–Meier). Ratio = Normal / Emergency; values <1 indicate Emergency yields higher per-turn discovery; values >1 indicate Normal yields higher per-turn discovery. Log-scale rendering (Fig. 1D in main paper) makes the crossover directly perceptible without numerical estimation.

Figure 3 plots the ratio on a log scale, reproducing the View D panel of Fig. 1 as a standalone figure for reference.

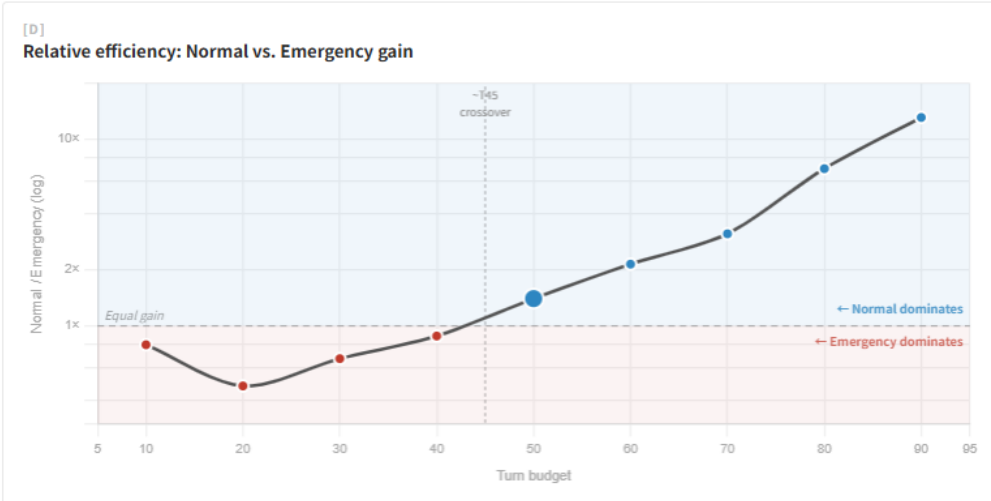

Figure 3: Normal/Emergency gain ratio on log scale (reproduced from Fig. 1D of main paper). Red points: Emergency dominates (ratio <1); blue points: Normal dominates (ratio >1). Dashed reference line at ratio = 1; crossover at ∼Turn 45. By Turn 90 the ratio reaches 13.1×.

H VIP Simulation Details

The Visually-Informed Protocol (VIP) applies a binary decision at Turn 30: sessions with at least one keyword detected by Turn 30 terminate at turn 30; remaining sessions extend to turn 100.

Coverage computation. Final coverage per group g is:

$$Cov_g = T30_g \cdot 1 + (1 - T30_g) \cdot T100_g$$

where $T30_g$ and $T100_g$ denote cumulative detection rates at Turn 30 and Turn 100 respectively (Table 4).

Average turns.

$$\bar{t}_{VIP} = \frac{\sum_g N_g(T30_g \cdot 30 + (1 - T30_g) \cdot 100)}{\sum_g N_g} = 47.9 \; turns$$

Table 7: VIP coverage by crisis level

Crisis Level	N	T30 (%)	T100 (%)	VIP Cov. (%)
Emergency	672	87	99	99
Abuse-Suspected	644	83	99	99
Counseling	621	78	98	98
Observation	634	66	94	94
Normal	665	58	94	94
Weighted avg	3,236	–	–	**96.8**

Note. VIP avg. turns = 47.9 t (vs. Oracle 67.8 t, Uniform-50 50.0 t). Gap recovery = $(94 - 58)/(100 - 58) = 85.7\%$ for Normal.

Threshold sensitivity. The T30-rule is robust across alternative cutoffs. Table 8 shows that Normal coverage (94%) and gap recovery (85.7%) are invariant across T20–T40; T30 minimizes average turns at 47.9 turns, confirming it as the efficiency optimum identified by View D's gain-ratio crossover.

Table 8: VIP sensitivity to threshold choice (T20–T40)

Threshold	Normal cov.	Avg t	Gap rec.
T20	94%	49.4 t	85.7%
T25	94%	48.2 t	85.7%
T30	**94%**	**47.9 t**	**85.7%**
T35	94%	49.2 t	85.7%
T40	94%	51.2 t	85.7%

Note. Coverage and gap recovery are invariant across all thresholds; T30 achieves minimum average turns.